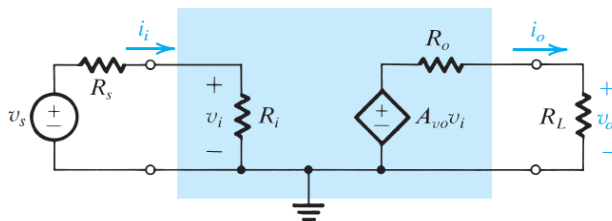


Homework 2

Topic: Amplifiers

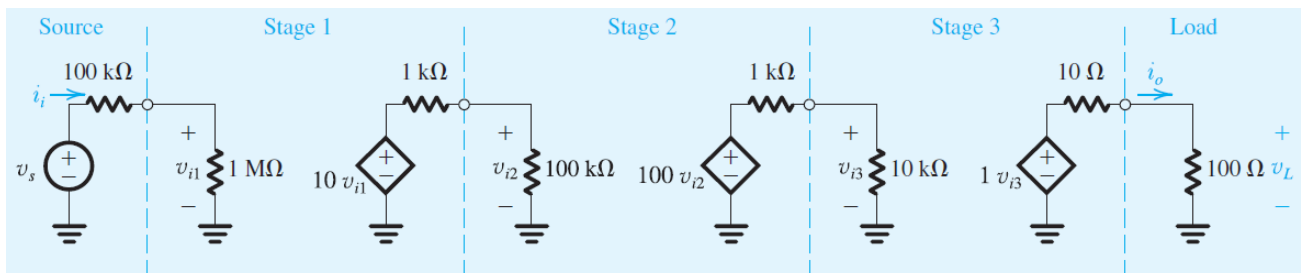
A1. For the circuit below where $A_{vo}=100$ V/V, calculate voltage gain, v_o/v_s , for the following three conditions (express your answers both directly and in decibels):

- (a) $R_i=10R_s$, $R_L=10R_o$
- (b) $R_i=R_s$, $R_L=R_o$
- (c) $R_i=R_s/10$, $R_L=R_o/10$



A2. An amplifier with 40 dB of small-signal, open-circuit voltage gain, an input resistance of 1 M Ω , and an output resistance of 100 Ω , drives a load of 500 Ω . What voltage and power gains (expressed in dB) would you expect with the load connected? If the amplifier has a peak output-current limitation of 20 mA, what is the rms value of the largest sine-wave input for which an undistorted output is possible? What is the corresponding output power available?

A3. Find the overall voltage gain v_o/v_s and compare to the overall voltage gain obtained when the first and second stages are interchanged.



A4. You are given two amplifiers, A and B, to connect in cascade between a 10-mV, 100-k Ω source and a 100- Ω load. The amplifiers have voltage gain, input resistance, and output resistance as follows: for A, 100 V/V, 100 k Ω , 10 k Ω , respectively; for B, 10 V/V, 10 k Ω , 1 k Ω , respectively. Your problem is to decide how the amplifiers should be connected. To proceed, evaluate the two possible connections between source S and load L, namely, SABL and SBAL. Find the voltage gain for each both as a ratio and in decibels. Which amplifier arrangement is best?

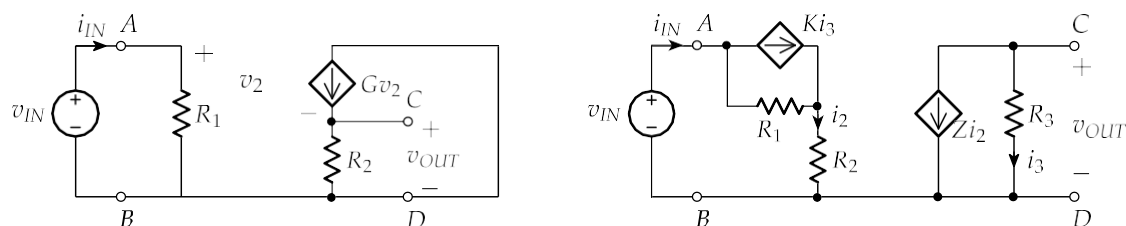
A5. Design an amplifier that provides 0.5W of signal power to a 100 Ω load resistance. The signal source provides a 30-mV rms signal and has a resistance of 0.5M Ω . Three types of voltage-amplifier stages are available:

- (a) A high-input-resistance type with $R_i=1$ M Ω , $A_{vo}=10$, and $R_o=10$ k Ω
- (b) A high-gain type with $R_i=10$ k Ω , $A_{vo}=100$, and $R_o=1$ k Ω
- (c) A low-output-resistance type with $R_i=10$ k Ω , $A_{vo}=1$, and $R_o=20$ Ω

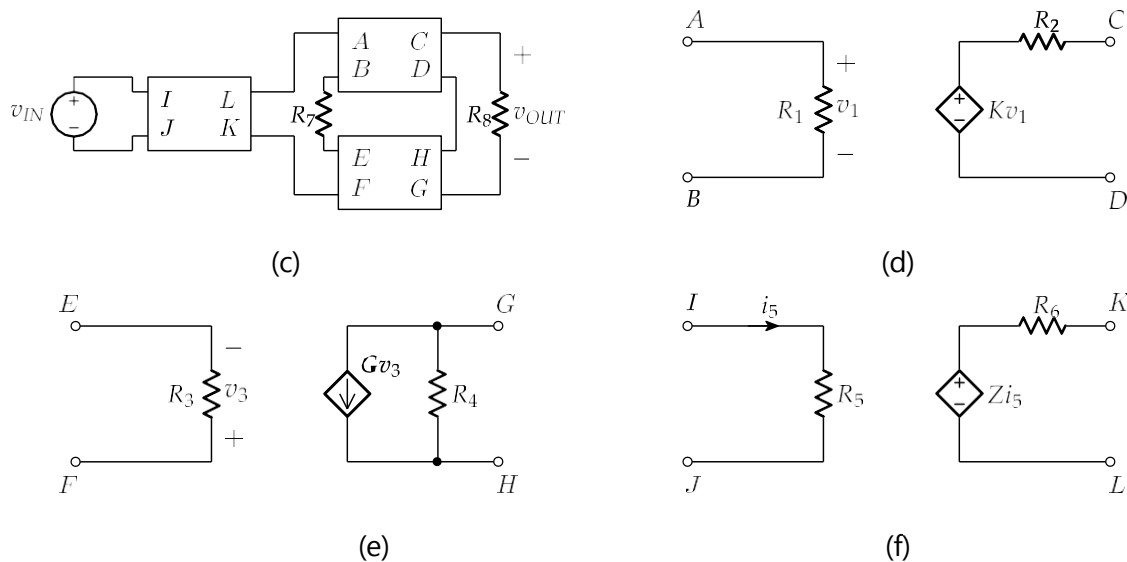
Design a suitable amplifier using a combination of these stages. Your design should utilize the minimum number of stages and should ensure that the signal level is not reduced below 10 mV at any point in the amplifier chain. Find the load voltage and power output realized.

A6. A transconductance amplifier with $R_i = 2 \text{ k}\Omega$, $G_m = 60 \text{ mA/V}$, and $R_o = 20 \text{ k}\Omega$ is fed with a voltage source having a source resistance of $1 \text{ k}\Omega$ and is loaded with a $1\text{-k}\Omega$ resistance. Find the voltage gain realized.

A7. For the following circuits, find the voltage gain, v_o/v_{IN} , and the input and output resistances.



A8. For the cascaded amplifier circuit in (c), find voltage gain, v_o/v_{IN} , and the input and output resistances using the circuits in (d)-(f).



A9. For the circuit below show

$$\frac{v_c}{v_b} = \frac{-\beta R_L}{r_\pi + (\beta + 1)R_E} \quad \text{and} \quad \frac{v_e}{v_b} = \frac{R_E}{R_E + [r_\pi / (\beta + 1)]}$$

